AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 8 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- Weak acid + NaOH: **K_a = [H^+] × salt ÷ acid**, where salt = mol NaOH added and acid = mol HA left
- Strong acid + strong base: find the excess mol of H^+ or OH^-, then divide by the **total** volume in dm^3
- Indicator: its colour-change range must lie within the vertical section of the pH curve
- pH = $-\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$

Section A: Short-answer questions

01 A 0.10 mol dm^{-3} aqueous solution of an acid is added slowly to 25 cm^3 of a 0.10 mol dm^{-3} aqueous solution of a base. Which acid–base pair has the highest pH at the equivalence point?

[1 mark]

A CH_3COOH and NaOH
B CH_3COOH and NH_3
C HCl and NaOH
D HCl and NH_3

02 Calculate the pH of 0.0350 mol dm^{-3} hydrochloric acid. Give your answer to 2 decimal places.

[1 mark]

03 Calculate the pH of a 0.150 mol dm^{-3} solution of ethanoic acid at 25 °C. Give your answer to 2 decimal places. For ethanoic acid, $K_a = 1.74 \times 10^{-5}$ mol dm^{-3} at 25 °C

[3 marks]

04 The table shows the pH ranges of some indicators.

Indicator	pH range
Bromocresol green	3.8 – 5.4
Bromothymol blue	6.0 – 7.6
Thymol blue	8.0 – 9.6

Identify the indicator that is most suitable for use in a titration between propanoic acid and sodium hydroxide.

[1 mark]

05 Equal volumes of two solutions, each with the same concentration, are mixed together at 298 K. Which two solutions, when mixed, form a solution with a pH > 7?

[1 mark]

A HCOOH and HCOOK
B KOH and CH_3COOH
C NH_3 and HCl
D NH_4Cl and KCl

Section B: Longer questions

06 25.0 cm^3 of 0.100 mol dm^{-3} ethanoic acid is titrated with 0.100 mol dm^{-3} sodium hydroxide at 25 °C.
For ethanoic acid, $K_a = 1.74 \times 10^{-5}$ mol dm^{-3}; $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6}

06.1 Calculate the pH of the ethanoic acid before any sodium hydroxide is added. Give your answer to 2 decimal places.

[2 marks]

06.2 Calculate the pH after 10.0 cm^3 of sodium hydroxide has been added. Give your answer to 2 decimal places.

[3 marks]

06.3 State the pH after 12.5 cm^3 of sodium hydroxide has been added. Explain your answer.

[1 mark]

06.4 Calculate the pH after 30.0 cm^3 of sodium hydroxide has been added. Give your answer to 2 decimal places.

[3 marks]

07 For pure water at 40 °C, pH = 6.67

A student thought that the water was acidic. Explain why the student was incorrect.

Determine the value of K_w at this temperature.

[4 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**A**	*Source: 7405/3 Nov 2020 Q14*	1
02	1.46	*Source: Practice question (written for this set)*	1
03	M1 $K_a = [H^+]^2 \div [CH_3COOH]$ M2 $[H^+] = \sqrt{(1.74 \times 10^{-5} \times 0.150)} = 1.62 \times 10^{-3}$ M3 pH = 2.79	*Source: 7405/1 Jun 2022 Q04.1* Answer must be to 2 dp	3
04	Thymol blue	*Source: 7405/1 Jun 2023 Q10.1*	1
05	**B**	*Source: 7405/3 Jun 2022 Q15*	1
06.1	M1 $[H^+] = \sqrt{(1.74 \times 10^{-5} \times 0.100)} = 1.32 \times 10^{-3}$ M2 pH = 2.88	*Source: Practice question (written for this set)*	2
06.2	M1 $n(CH_3COOH)$ left $= 2.50 \times 10^{-3} - 1.00 \times 10^{-3} = 1.50 \times 10^{-3}$ mol M2 $n(CH_3COO^-) = 1.00 \times 10^{-3}$ mol M3 $[H^+] = 1.74 \times 10^{-5} \times 1.50 \div 1.00 = 2.61 \times 10^{-5}$; pH = 4.58		3
06.3	4.76 — half-neutralised so $[HA] = [A^-]$ and pH = pK_a		1
06.4	M1 excess $n(OH^-) = 3.00 \times 10^{-3} - 2.50 \times 10^{-3} = 5.0 \times 10^{-4}$ mol M2 $[OH^-] = 5.0 \times 10^{-4} \div 0.0550 = 9.09 \times 10^{-3}$ M3 pH = 11.96	M2 must use total volume 55.0 cm^3	3
07	M1 $[H^+] = [OH^-]$ M2 $[H^+]$ $(= 10^{-pH}) = 2.138 \times 10^{-7}$ M3 $K_w = [H^+]^2$ or $(2.138 \times 10^{-7})^2$ M4 $K_w = 4.57 \times 10^{-14}$ ($mol^2\ dm^{-6}$)	*Source: 7405/1 Jun 2019 Q09.1* M1 accept equal number / amounts of H^+ and OH^- M4 allow 4.58×10^{-14}; M4 dependent on squaring in M3	4